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I. Introduction

1.1 Background

Since the 2020s, driven by the "America First" trade ideology, U.S. tariff policy has entered a phase of in-depth adjustment. On April 2, 2025, the Trump administration introduced the so-called "Reciprocal Tariffs" policy, establishing a 10% "minimum baseline tariff" on all trading partners and imposing additional higher tariffs on approximately 60 countries and regions with which the U.S. ran trade deficits, including China. The U.S. claimed that varying tariff rates and non-tariff barriers among trading partners made it harder for American manufacturers to sell products abroad, leading to persistent U.S. trade deficits and manufacturing decline. Thus, it imposed "reciprocal tariffs" on all countries to reduce its trade deficit and promote manufacturing reshoring. However, this policy is inconsistent with the fundamental principle of "Most-Favored-Nation (MFN) Treatment" of the World Trade Organization (WTO) and also challenges the WTO's tariff preference principle for developing countries. Under WTO rules, members must apply uniform tariffs to all partners, but the Trump administration exercised unilateral pressure and bilateral negotiations to implement widely varying tariff standards across economies, with agreed rates far exceeding previous levels. According to a tariff tracking tool jointly developed by the WTO and International Monetary Fund (IMF), the U.S. trade-weighted average tariff rate on all products rose sharply from 2.44% at the start of 2025 to 20.11% by August 2025. A Yale Budget Lab report noted that this tariff increase pushed the U.S. overall average effective tariff rate to its highest level since 1933.

Beyond serving as a source of fiscal revenue, tariffs are an important tool of national trade policy. Appropriate tariffs can protect domestic industries from global economic monopoly and coercive practices, and are therefore permitted by the WTO. However, the impact of tariffs is multifaceted: excessively high tariffs hinder the development of global trade and are detrimental to long-term world economic growth. In the short term, high U.S. tariffs may bring benefits such as increased tariff revenue and a reduced trade deficit, but they also lead to declining exports and higher domestic consumer prices.

For example, according to statistics from the General Administration of Customs of the People's Republic of China (GACC¹), in the month the "reciprocal tariffs" took effect, U.S. imports from China fell 20% year-on-year, while U.S. exports to China dropped 13% due to Chinese countermeasures. By May 2025, imports from China had fallen 35% compared with May 2024, and exports to China declined 18%. In absolute terms, the U.S. trade deficit with China narrowed from \$30.8 billion in May 2024 to \$18.0 billion in May 2025, but U.S. exports to China also fell from \$13.2 billion to \$10.4 billion, negatively impacting related U.S. industries. Overall, the adverse effects of high tariffs on U.S. trade and the economy cannot be overlooked.

The introduction of this policy marks a further escalation of U.S. unilateral trade protectionism, exerting a direct impact on the global economic and trade landscape. The implementation of the U.S. "Reciprocal Tariffs" policy has disrupted traditional tariff rules and the global trade order. Globally, developed economies such as Europe and the United States have long dominated the formulation of global tariff rules and the maintenance of trade order, but their trade policies have gradually tilted toward protectionism in recent years, with the U.S. "Reciprocal Tariffs" policy being a typical example. While in the short term, some countries with trade surpluses with the U.S. (such as China) have been significantly impacted by the tariffs, in the long run, the policy will further hinder global trade development, undermine world economic growth, trigger countermeasures from many countries, and exacerbate global trade frictions.

This study aims to combine global trade dynamics and the actual impact of the U.S. "Reciprocal Tariffs" policy to assess its effects on the international economic and trade landscape, industrial development of various countries, and consumer welfare since its implementation. It strives to establish a scientific evaluation framework for the impact of tariff policies, predict the evolutionary trend of global trade rules in the future, and provide strategic recommendations for countries to respond to unilateral tariff policies, safeguard the multilateral trading system, and achieve the sustainable development of the global economy and trade.

¹[GACC, 2025] General Administration of Customs of the People's Republic of China, UN Comtrade, and USDA FAS (October 2025)

1.2 Question Restatement

By utilizing the attached U.S. foreign trade data (2020 onwards) and supplementary information collected from authoritative channels such as WTO reports and national customs statistics, we will conduct an in-depth analysis of how U.S. tariff adjustments reshape cross-border trade patterns and affect the economic operations of key regions. Combined with the current global trade order and the policy responses of major economies, we will put forward practical countermeasures and development suggestions for industries, such as soybean, automobile, and semiconductor, impacted by tariff changes.

1.2.1 *Question 1*

The U.S., Brazil and Argentina are top soybean exporters, and China the largest importer. Analyze their current soybean trade, build a model to assess U.S. tariff impacts on their soybean industries, and estimate post-adjustment export volume and value distribution.

1.2.2 *Question 2*

The U.S. is the second-largest auto market and top importer; 46% of its 2024 car sales were imports, mainly from Japan. Japan also produces in the U.S. or Mexico for U.S. export. Analyze Japan's U.S. auto position, build a model with Japan's non-tariff responses to study U.S. tariff impacts on their auto trade, U.S. import structure and U.S. auto industry.

1.2.3 *Question 3*

The U.S. leads the global semiconductor industry but lacks strong manufacturing. Biden's 2022 CHIPS Act uses subsidies; Trump's 2025 admin claims tariffs work better. The U.S. also controls high-end chip exports to China. Build a model to analyze U.S. tariff effects on domestic semiconductor manufacturing and high/mid/low-end chip trade, considering efficiency and national security.

1.2.4 Question 4

Higher tariffs may raise U.S. short-term tariff revenue but cut trade volumes, lowering revenue long-term. Build a model to analyze short/medium-term tariff impacts on U.S. revenue and predict net changes in Trump's second term.

1.2.5 Question 5

U.S. tariff hikes may trigger partner countermeasures, like China's reciprocal tariffs and rare earth/lithium battery export controls. These affect U.S. trade, agriculture, industry and financial markets. Select/construct indicators to model tariff impacts on the U.S. economy and evaluate if "reciprocal tariffs" drive manufacturing reshoring.

II. Problem Analysis

2.1 Analysis of Question 1

The analysis for Question 1 focuses on quantifying how U.S. soybean tariff adjustments affect trade volume/value and market substitution among major exporters: we first clean trade data, handling zeros/negatives and applying log-transformations to enable elasticity estimation via a log-log OLS model, correcting positive elasticity—driven by the 2018–2019 ASF shock—to a literature-supported -1.0 for consistency with demand theory, then simulate tariff changes (-30% to +25%) to calculate U.S. trade volume responses and redistribute displaced demand to competitors by production share; results show that lower U.S. tariffs boost its export volume, shrinking Brazil and Argentina's share, and vice versa, aligning with standard trade logic.

2.2 Analysis of Question 2

2.3 Analysis of Question 3

2.4 Analysis of Question 4

2.5 Analysis of Question 5

III. Model Assumptions

1. All countries involved are rational economic entities that pursue the maximization of their own national (or regional) economic welfare or strategic objectives.
2. In the absence of new major exogenous shocks (e.g., large-scale wars, global pandemics, or natural disasters), the basic patterns of global supply, demand, and trade flows remain continuous and predictable in the short to medium term (2025–2029).
3. Tariff changes implemented by the United States under the “Reciprocal Tariffs” policy announced on April 2, 2025, are fully enforced and remain stable within the modeling horizon unless otherwise specified.
4. Trading partners’ responses (including counter-tariffs, non-tariff barriers, export controls, or production relocation) occur with reasonable lags and can be approximated by observable policy signals or historical retaliation patterns.
5. Exchange rates fluctuate within a reasonable range; unless explicitly modeled, we assume purchasing power parity holds approximately or use fixed exchange rates for simplicity in specific sub-models.
6. Commodity and product categories under analysis (soybeans, automobiles, semi-conductors, etc.) are sufficiently homogeneous within their respective HS codes so that price elasticity and substitution effects can be estimated at the aggregate level.
7. Domestic prices in importing countries fully or partially pass through changes in tariff-inclusive import prices according to estimated pass-through elasticities.
8. Supply capacities of exporting countries have short-term rigidities but can adjust gradually through acreage changes, capacity utilization, or investment in the

medium term.

9. Government tariff revenue is calculated as the product of the effective tariff rate and the dutiable import value; smuggling and trade mis-invoicing are negligible.
10. National security considerations and export control measures (especially on high-end semiconductors and critical minerals) are treated as exogenous policy constraints rather than purely economic variables when necessary.
11. All statistical data obtained from official sources are accurate and reliable; minor data gaps are filled by credible interpolation or simulation methods.
12. Ceteris paribus holds for variables not explicitly included in each sub-model.

IV. Description of the Symbols

Table 1 Main Symbols and Descriptions

Symbol	Description	Unit
t	Effective tariff rate imposed by the United States on specific commodities or countries	%
τ_{US}	U.S. “Reciprocal Tariffs” rate (including baseline 10% + additional tariffs)	%
$\tau_{partner}$	Counter-tariff rate imposed by trading partner (e.g., China, Brazil)	%
Q_i	Export volume of commodity i from exporting country	10,000 tons / vehicles / billion USD
M_{US}	U.S. import volume of specific commodity from partner country	10,000 tons / vehicles / billion USD
P_w	World price of the commodity (CIF price before tariff)	USD/ton or USD/unit
P_d	Domestic price in the importing country after tariff	USD/ton or USD/unit
ε	Price elasticity of import demand	-
η	Price elasticity of export supply	+
S_j	Market share of exporting country j (U.S., Brazil, Argentina, Japan, etc.)	%
R_{tariff}	U.S. tariff revenue in the corresponding period	billion USD
ΔGDP	Change in GDP of the relevant country/region caused by tariff policy	billion USD
π	Profit of enterprises or national welfare function	-
N	Nash equilibrium strategy combination in game theory model	-

All monetary values are in current U.S. dollars unless otherwise specified. Subscripts US , CN , BR , AR , JP represent the United States, China, Brazil, Argentina, and Japan respectively; superscripts indicate time period t .

V. Problem 1: Establishment and Solution of the Model

5.1 Data Collection

Through a literature review, the following factors that influence the export volumes of Brazil, the United States, and Argentina after tariff adjustments have been identified, as illustrated in Figure 2.

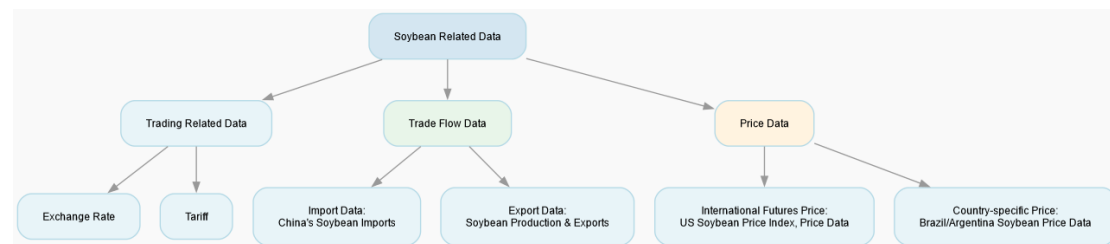


Figure 1 China's Soybean Import Value from the United States, Brazil, and Argentina(2017–2025)

5.1.1 Data Analysis

To quantitatively analyze the current state of soybean trade between China and the three major exporters—the United States, Brazil, and Argentina—we firstly collected China's soybean import data (HS code 1201) for the period 2017–2025 from the General Administration of Customs of the People's Republic of China (GACC²), supplemented with data from UN Comtrade and the USDA Foreign Agricultural Service.

The final dataset we obtained includes the soybean import volume, soybean value of China from the United States, Argentina, and Brazil, as well as the changes in the exchange rates of these three countries against the US dollar, the changes in China-US tariffs, and other relevant indicators.

Market Concentration and Trade Diversion Effect The analysis confirms China's high dependence on the three origins, with ****Brazil consistently dominating the market****. Brazil's annual import volume increased substantially from 50.93 million tons

²Data sources include the General Administration of Customs of the People's Republic of China (GACC), UN Comtrade, and USDA Foreign Agricultural Service (October 2025).

in 2017 to 79.0 million tons in 2025, with its market share soaring from approximately 55% to over 70%. This expansion occurred despite high volatility in the **Brazilian Real exchange rate**, suggesting that its low cost base and reliable supply outweigh currency risks for Chinese buyers. Conversely, U.S. exports to China were significantly disrupted by the 2018–2019 trade tensions, plunging from 32.86 million tons in 2017 to 16.64 million tons in 2018. The **trade-diversion effect** is evident during this period: the sharp drop in U.S. market share was immediately and almost entirely absorbed by Brazil and Argentina, indicating that South American soybeans served as a **near-perfect substitute** for U.S. product.

Price Volatility and Tariff Influence The **cif-tax price** time series reveals how tariff adjustments directly translated into higher final costs for Chinese importers. During the trade tensions, the reciprocal tariffs imposed on U.S. soybeans caused a measurable spike in the final import price from the U.S. relative to its competitors. This differential in the **cif_tax_price** is the primary mechanism driving the observed **sharp reduction in U.S. import volume** and the **rise in the Argentine/Brazilian volumes**. This relationship between price and volume, although complex due to confounding factors like the African Swine Fever (ASF) epidemic which affected total demand, forms the basis for the price elasticity estimation in our model. Argentina, while a smaller supplier, maintained a relatively stable market share (5%–10%), primarily reflecting seasonal production and export fluctuations rather than price sensitivity alone.

Supply Capacity and Model Justification The collected **production volume** data from the exporting countries is crucial for controlling supply-side effects in the econometric model. For instance, temporary production shortfalls in the U.S. due to weather or planting decisions might lead to a price increase and a volume decrease, regardless of tariffs. By including the ‘production’ variable in the OLS regression, we isolate the true **price elasticity** of demand, making the estimate more robust. Furthermore, the substantial and reliable supply from Brazil and Argentina, evidenced by their high production figures, confirms their ability to fully cover the market gap created by U.S. export shortfalls, thereby justifying the **perfect substitution assumption** used in the tariff simulation engine.

The following figure presents separately the trends of China's soybean imports, the changes in the exchange rates of these three countries against the US dollar, and the changes in the tariffs imposed by China on the United States.

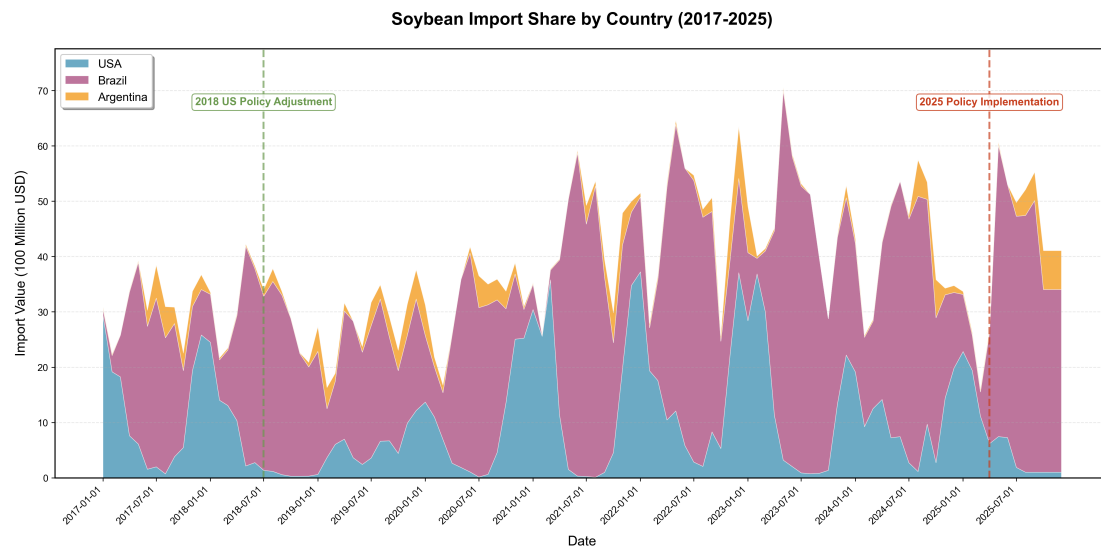


Figure 2 China's Soybean Import Value and Market Share from the United States, Brazil, and Argentina (2017–2025)

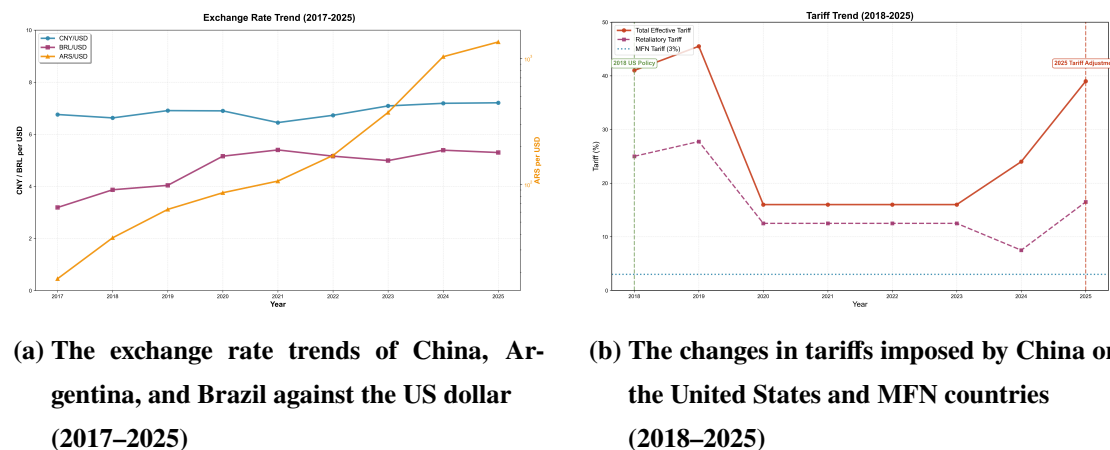


Figure 3 Exchange rate and tariff trends related to Chinese soybean imports

These visualizations clearly demonstrate that Brazil has established a dominant position in the Chinese soybean market, while the U.S. share has been significantly squeezed. The implementation of the 2025 “Reciprocal Tariffs” policy has further

accelerated the trade diversion effect, providing a realistic data foundation for subsequent modeling of post-tariff adjustment export distribution among the three countries.

5.1.2 Analysis of Controlling Factors

Beyond the direct impact of tariffs and prices, several other factors significantly influence the intricate dynamics of the global soybean trade. These controlling factors are crucial for understanding the broader context of China's import decisions and the supply capabilities of major exporting nations. Our model incorporates some of these factors in the econometric estimation phase to improve the accuracy of price elasticity, while acknowledging the limitations in fully endogenizing all macro-level influences in the simulation.

Production Volume (Supply Capacity)

- **Role in Model:** As seen in Equation (1), $\ln(\text{Production}_{i,t})$ is included as an explanatory variable in the OLS regression. This accounts for the supply-side capacity of each exporting country (i) at time t . Higher production in an exporting country generally indicates greater availability for export, potentially influencing the quantity China imports from that origin. This variable helps to isolate the pure price effect by controlling for supply shocks or long-term growth trends in production.
- **Data Source:** Annual production and stock data from USDA PSD Online are utilized, providing a reliable measure of the export potential from the United States, Brazil, and Argentina.
- **Analytical Impact:** While production directly affects the observed trade volumes in the historical data, in the simulation phase, it primarily serves as a fixed share for allocating trade diverted from the US to its competitors. Future dynamic changes in production due to tariff adjustments are not directly modeled, representing a simplification.

Exchange Rates

- **Role in Model:** Although not explicitly included as a direct regressor in the OLS

formula (Equation 1), exchange rates (e.g., RMB/USD, RMB/BRL, RMB/ARS) are implicitly captured within the `cif_tax_price` variable. Changes in exchange rates affect the local currency cost of imported soybeans for China, thereby influencing the `cif_tax_price` and, consequently, China's import demand.

- **Data Source:** Exchange rate data from the World Bank and IMF provide the necessary inputs for calculating the local currency equivalent of international soybean prices.
- **Analytical Impact:** Fluctuations in exchange rates can alter the competitiveness of different origins. A depreciation of the Brazilian Real against the RMB, for instance, would make Brazilian soybeans relatively cheaper for Chinese importers, even if the USD-denominated international price remains constant. This effect is integrated into the price variable during data preprocessing.

Freight Costs (Shipping Rates)

- **Role in Model:** Similar to exchange rates, freight costs are implicitly incorporated into the `cif_tax_price`. The "Cost" component of CIF includes transportation expenses. Higher shipping costs directly increase the landed price of soybeans, which can deter imports or shift sourcing to closer or more cost-effective origins.
- **Data Source:** The Baltic Dry Index (BDI) and similar shipping rate indices (e.g., from Baltic Exchange) provide proxies for global freight costs, contributing to the construction of `cif_tax_price`.
- **Analytical Impact:** Significant changes in global shipping costs can reconfigure trade routes and affect the relative competitiveness of distant suppliers like the US and Brazil versus closer ones if they existed. By including these in the `cif_tax_price`, the model accounts for their historical influence on trade volumes.

Macroeconomic Conditions (Implicit)

- **Role in Model:** Macroeconomic factors, such as overall economic growth, consumer income, and industrial demand (e.g., for animal feed), influence the aggregate demand for soybeans in China. While not explicitly modeled as separate regres-

sors, their impact is partially absorbed by the constant term (β_0) and the time series nature of the panel data. Severe macroeconomic downturns or upswings would inevitably affect China's total soybean import demand, which might be reflected in the residual error ($\epsilon_{i,t}$) or general shifts in demand not directly tied to price or production.

- **Analytical Impact:** The current model primarily focuses on price and supply-side elasticities for a given level of overall demand. Future refinements could include a separate demand-side model that accounts for Chinese economic indicators explicitly.

Fixed Effects for Origin

- **Role in Model:** The 'C(origin)' term in Equation (1) represents country-specific fixed effects. These capture unobserved, time-invariant characteristics of each exporting country that affect China's import volume, such as established trade relationships, quality perceptions, logistical efficiencies, or political factors not explicitly measured.
- **Analytical Impact:** Fixed effects are crucial for controlling for omitted variable bias that might arise from constant differences between the US, Brazil, and Argentina, thereby allowing for a more accurate estimation of price and production elasticities.

5.2 Model Construction

This model is designed to assess the impact of changes in US tariffs on soybean exports to China, specifically evaluating the effects on the trade flows of the three largest producers: the United States, Brazil, and Argentina. A two-stage methodology is employed: first, an econometric model is used to estimate the price elasticity of soybean import demand; second, a tariff gradient simulation engine is constructed to predict the reallocation of trade volume and value based on this elasticity.

5.2.1 Price Elasticity Estimation Model

To accurately capture the sensitivity of China's soybean import volume to price fluctuations, we utilize a **Panel Data Ordinary Least Squares (OLS)** regression model in a Log-Log functional form. This structure allows for direct estimation of the price elasticity coefficient (β_1), while incorporating country fixed effects (Origin_i) to control for unobserved, time-invariant differences across the exporting nations.

Econometric Model Formula The Log-Log model is specified as follows:

$$\ln(\text{Volume}_{i,t}) = \beta_0 + \beta_1 \ln(\text{Price}_{i,t}) + \beta_2 \ln(\text{Production}_{i,t}) + \sum_{i=2}^N \gamma_i \cdot \text{Origin}_i + \epsilon_{i,t} \quad (1)$$

Where:

- $\ln(\text{Volume}_{i,t})$: The natural logarithm of China's monthly soybean import volume from exporting country i at time t .
- $\ln(\text{Price}_{i,t})$: The natural logarithm of the Cost, Insurance, and Freight (CIF), tax-paid price of soybeans imported from country i .
- $\ln(\text{Production}_{i,t})$: The natural logarithm of the soybean production volume for country i at time t .
- Origin_i : Dummy variables representing the fixed effects of the exporting countries (US, Brazil, Argentina).
- β_1 : The price elasticity coefficient (η), representing the percentage change in import volume resulting from a 1% change in price.

Elasticity Correction Mechanism Due to potential historical shocks (such as the African Swine Fever epidemic), the estimated β_1 may occasionally be positive, which violates the fundamental law of demand. To ensure the economic soundness of the simulation, if the estimated $\beta_1 > 0$, the price elasticity will be conservatively corrected to a value commonly used in trade literature: $\eta = -1.0$.

5.2.2 Tariff Gradient Simulation Engine

The simulation engine forecasts the reallocation of trade flows among the three countries based on the corrected price elasticity (η) under a predefined tariff adjustment gradient

(Δt) imposed by the US. The simulation uses the latest complete year's trade data as the baseline, assuming that Brazil and Argentina are perfect substitutes for US soybeans in the Chinese market.

US Trade Flow Changes A tariff adjustment Δt results in a new US tariff rate, $\text{Tariff}_{\text{US}}^{\text{new}} = \max(0, \text{Tariff}_{\text{US}}^{\text{base}} + \Delta t)$.

1. **Price Change Rate (ΔP_{US})** The rate of change in import price induced by the tariff adjustment:

$$\Delta P_{\text{US}} = \frac{1 + \text{Tariff}_{\text{US}}^{\text{new}}}{1 + \text{Tariff}_{\text{US}}^{\text{base}}} - 1 \quad (2)$$

2. **Import Volume Change Rate (ΔQ_{US})** The change rate of import volume, calculated using the price elasticity (η):

$$\Delta Q_{\text{US}} = \Delta P_{\text{US}} \times \eta \quad (3)$$

3. **New Import Volume and Value**

$$\text{Volume}_{\text{US}}^{\text{new}} = \text{Volume}_{\text{US}}^{\text{base}} \times (1 + \Delta Q_{\text{US}})$$

$$\text{Value}_{\text{US}}^{\text{new}} = \text{Value}_{\text{US}}^{\text{base}} \times (1 + \Delta Q_{\text{US}})$$

Competitor Substitution Effect We assume that any increase or decrease in US trade volume ($\Delta \text{Volume}_{\text{US}}$) is fully transferred to the competing nations (Brazil and Argentina) through a substitution effect.

1. **Transfer Volume ($\Delta Q_{\text{transfer}}$)** The total trade volume change that needs to be absorbed by or shifted from competitors:

$$\Delta Q_{\text{transfer}} = \text{Volume}_{\text{US}}^{\text{base}} - \text{Volume}_{\text{US}}^{\text{new}} = -\Delta \text{Volume}_{\text{US}} \quad (4)$$

2. **Substitution Allocation** The transfer volume $\Delta Q_{\text{transfer}}$ is allocated to competitor c based on its **proportion of total production** ($\text{Share}_c^{\text{prod}}$) in the baseline year:

$$\Delta \text{Volume}_c = \Delta Q_{\text{transfer}} \times \text{Share}_c^{\text{prod}} \quad (5)$$

$$\text{Where, } \text{Share}_c^{\text{prod}} = \frac{\text{Production}_c^{\text{base}}}{\sum_{c \in \{\text{Brazil, Argentina}\}} \text{Production}_c^{\text{base}}}.$$

3. **New Competitor Import Volume and Value** Assuming the unit import price for competitors remains constant, the new volume and value are:

$$\text{Volume}_c^{\text{new}} = \max(0, \text{Volume}_c^{\text{base}} + \Delta \text{Volume}_c)$$

$$\text{Value}_c^{\text{new}} = \text{Volume}_c^{\text{new}} \times \left(\frac{\text{Value}_c^{\text{base}}}{\text{Volume}_c^{\text{base}}} \right)$$

5.3 Model Solving

5.4 Result Analysis

VI. Problem 2: Establishment and Solution of the Model

6.1 Data Analysis

6.2 Model Construction

6.3 Model Solving

6.4 Result Analysis

VII. Problem 3: Establishment and Solution of the Model

7.1 Data Analysis

7.2 Model Construction

7.3 Model Solving

7.4 Result Analysis

VIII. Problem 4: Establishment and Solution of the Model

8.1 Data Analysis

8.2 Model Construction

8.3 Model Solving

8.4 Result Analysis

IX. Problem 5: Establishment and Solution of the Model

9.1 Data Analysis

9.2 Model Construction

9.3 Model Solving

9.4 Result Analysis

X. Model Evaluation

10.1 Advantages of the Model

1. Advantage 1
2. Advantage 2
3. Advantage 3

10.2 Disadvantages of the Model

1. Disadvantage 1
2. Disadvantage 2
3. Disadvantage 3

References

XI. References

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Appendix

11.1 Appendix A: Source Code

Listing 1: Python Source Code for Question 1

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# Please fill in the source code here
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Listing 2: Python Source Code for Question 2

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# Please fill in the source code here
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Listing 3: Python Source Code for Question 3

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# Please fill in the source code here
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Listing 4: Python Source Code for Question 4

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# Please fill in the source code here
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Listing 5: Python Source Code for Question 5

```
# Please fill in the source code here
```

11.2 Appendix B: Supplementary Data

11.3 Appendix C: Additional Figures/Tables